

Tucson Crime Insights: Predicting and Understanding Crime Patterns Using Statistical Machine Learning

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1 Introduction

The Tucson Police Department publishes crime incident records through its open data portal, covering reported incidents across the city from 2018 onward. This project applies statistical machine learning methods to 243,127 cleaned records spanning 2018 through 2025 to investigate two questions: (1) Can we predict the type of crime from temporal and geographic features such as hour of day, day of week, division, ward, and how the crime was reported? (2) What natural groupings emerge from these same features when crime type labels are withheld?

The dataset presents several real-world challenges: severe class imbalance (Larceny accounts for approximately 70% of records while Homicide represents approximately 0.2%), inconsistent time formatting requiring custom parsing, and missing geographic identifiers, making it a realistic environment for evaluating classification and clustering pipelines. The motivation is local: Tucson is where I currently live and study, and the patterns uncovered have direct implications for public safety resource allocation.

2 Dataset Description

The dataset was obtained from the Tucson Police Department Open Data Portal and contains 251,280 records covering January 2018 through January 2026. Each record represents a single reported crime incident with 14 variables, described in Table 1.

Table 1: Dataset variable descriptions

Variable	Description
IncidentID	Unique identifier for each crime incident
DateOccurred	Full timestamp of when the crime occurred
Year	Year of occurrence (2018 to 2026)
Month	Month of occurrence
Day	Day of week (Mon to Sun)
TimeOccur	Time of occurrence; inconsistent formatting, required custom parsing
Division	Geographic police division (East, West, South, Midtown)
Ward	City council ward number 1 to 6; contains missing values
UCR	Numeric Uniform Crime Reporting category code
UCRDescription	Broad crime category label; primary target variable
Offense	Specific offense sub-code
OffenseDescription	Detailed offense label (e.g. Larceny Shoplifting)
CallSource	How the crime was reported (Call For Service, Officer Initiated, Web Reported, Detective)
ESRI_OID	Row index assigned by mapping software; not used in analysis

Three variables contain missing values in the raw data: Division (3,743 missing), Ward (15,566 missing), and CallSource (7,408 missing). The primary target variable is UCRDescription, which contains 8 broad crime

categories: Homicide, Sexual Assault, Robbery, Aggravated Assault, Burglary, Larceny, GTA (Grand Theft Auto), and Arson.

3 Data Exploration and Preprocessing

3.1 Data Cleaning

Several cleaning steps were applied to the raw dataset before analysis. Records from 2026 were dropped since only 23 days of data exist for that year, making year-level comparisons unreliable. Records with missing Division values (3,743 rows) were dropped since Division is a core location feature used in modeling. Records labeled Other Jurisdiction in the Division column (140 rows) were also removed as they represent crimes outside Tucson city limits. Missing Ward values (15,566 rows) were filled with 0 to indicate unknown rather than dropping those records, since Ward is a secondary feature and dropping them would cause significant data loss. Missing CallSource values (7,408 rows) were filled with Unknown for the same reason. After cleaning, the final dataset contains 243,127 records.

The most significant cleaning challenge was the TimeOccur column, which contained at least three distinct formats across records. Entries like 113, 0428, and 13:32 all represent valid times but in inconsistent formats. A custom parser was written to standardize all entries to a numeric hour (0 to 23). Entries beginning with 24 such as 2400 were treated as midnight (hour 0). All 243,127 records were successfully parsed with zero failures.

3.2 Exploratory Data Analysis

The following figures summarize the key patterns in the cleaned dataset across crime type, time, geography, and reporting method.

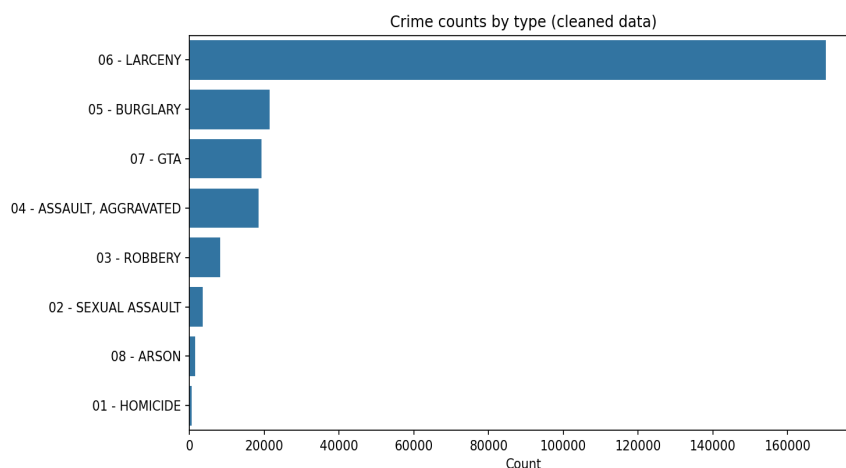


Figure 1: Crime counts by type in the cleaned dataset. Larceny accounts for approximately 70% of all records, creating severe class imbalance.

The class imbalance in this dataset is severe. Larceny dominates with 170,264 incidents while the next largest category, Burglary, has only 21,409. Homicide and Arson together account for fewer than 2,100 records across seven years. This imbalance fundamentally shapes every modeling decision in this project, from the choice of evaluation metric to how class weights are applied during training.

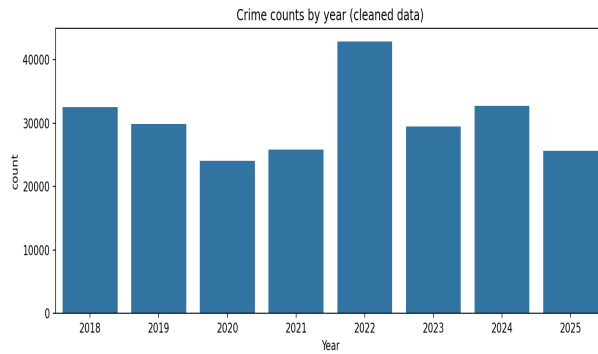


Figure 2: Annual crime counts (2018 to 2025). Crime peaked in 2022.

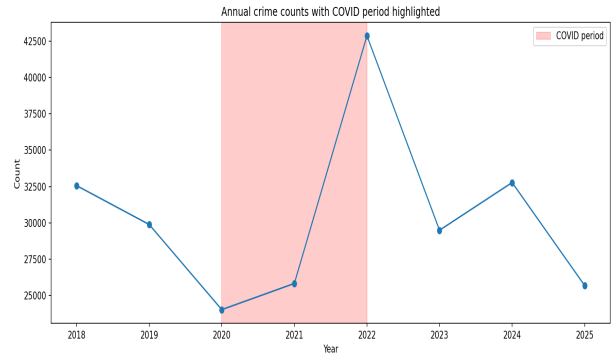


Figure 3: Annual counts with COVID period highlighted (shaded red).

Crime volume shows a clear COVID-19 effect. Counts dropped noticeably in 2020 and 2021 before recovering sharply and peaking at approximately 43,000 incidents in 2022. Larceny drove the largest absolute decline during this period, likely reflecting reduced retail activity during lockdowns, while violent crime categories remained comparatively stable.

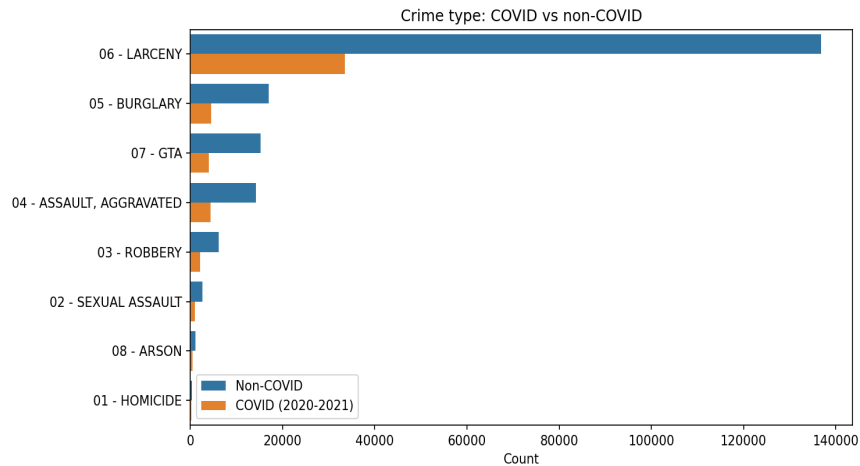


Figure 4: Crime type distribution during COVID (2020 to 2021) versus non-COVID years. Larceny saw the largest absolute decline.

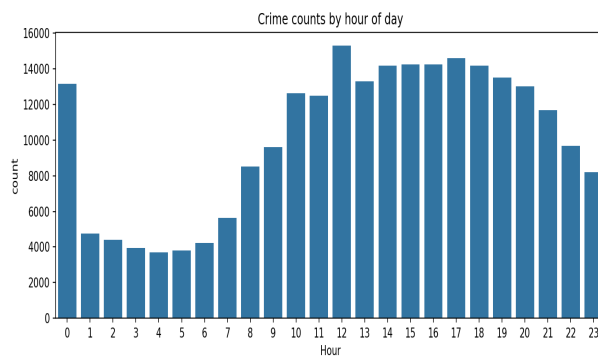


Figure 5: Crime counts by hour of day. Activity peaks around noon.

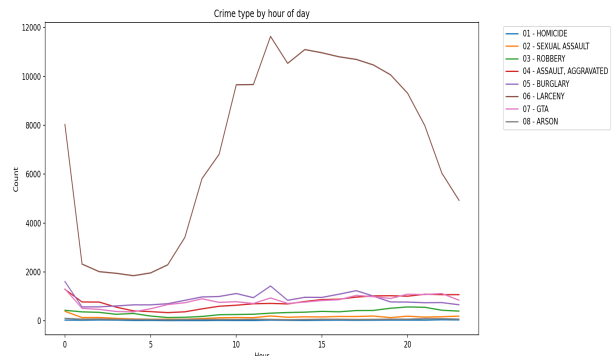


Figure 6: Crime type by hour. Each crime type has a distinct hourly profile.

When broken down by hour, different crime types reveal distinct temporal patterns. Larceny peaks sharply around noon and remains elevated through the afternoon, consistent with daytime retail activity. Aggravated Assault and GTA show a flatter distribution with higher evening and late-night counts. This variation in hourly patterns across crime types is why Hour becomes the single most important feature in the classification models.

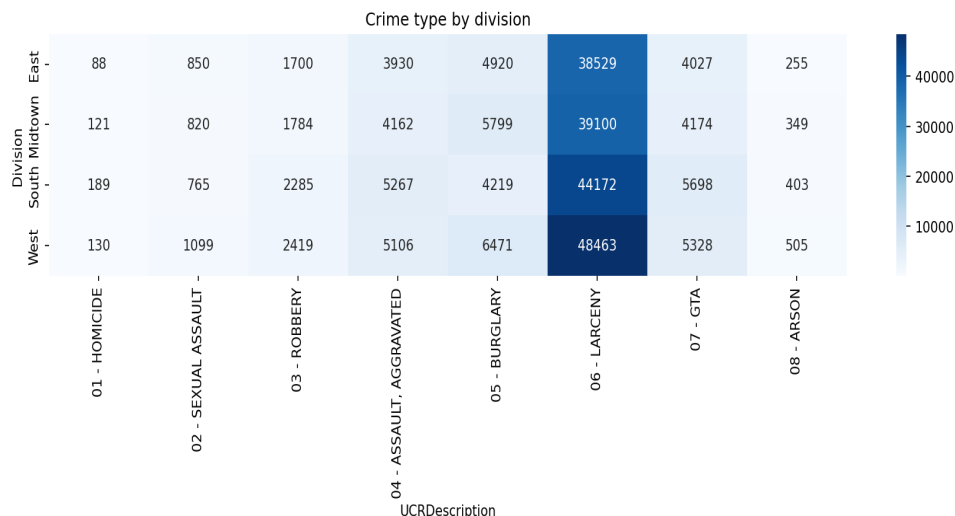


Figure 7: Heatmap of crime type counts by police division. Larceny dominates in every division; West Division has the highest overall volume.

Across all four police divisions, Larceny is the dominant crime type by a wide margin. West Division records the highest overall volume. The distribution of non-Larceny crimes is relatively consistent across divisions, suggesting that geographic division alone provides limited additional signal for distinguishing crime types beyond what is already captured by call source and time features.

4 Feature Selection

Before training classification models, feature selection was performed to identify which variables are most predictive of crime type. Three columns were dropped prior to modeling: IncidentID (a unique identifier with no predictive value), UCR (a numeric code that directly encodes the target variable UCRDescription and would constitute data leakage), and Offense (a sub-code that similarly maps one-to-one with the target). The remaining categorical features including Division, CallSource, Ward, Month, and Day were one-hot encoded using pandas get_dummies with drop_first=True to avoid multicollinearity, expanding the feature set to 32 columns.

Feature selection using L1 regularized logistic regression was initially attempted, but under default regularization strength it shrunk all but two features to zero coefficients, making it too aggressive for this imbalanced dataset. Random Forest feature importance was used instead, as it is more robust to class imbalance and does not require scaling. Features with an importance score below 0.01 were dropped, reducing the final feature set to 22 columns used across all classification and clustering models.

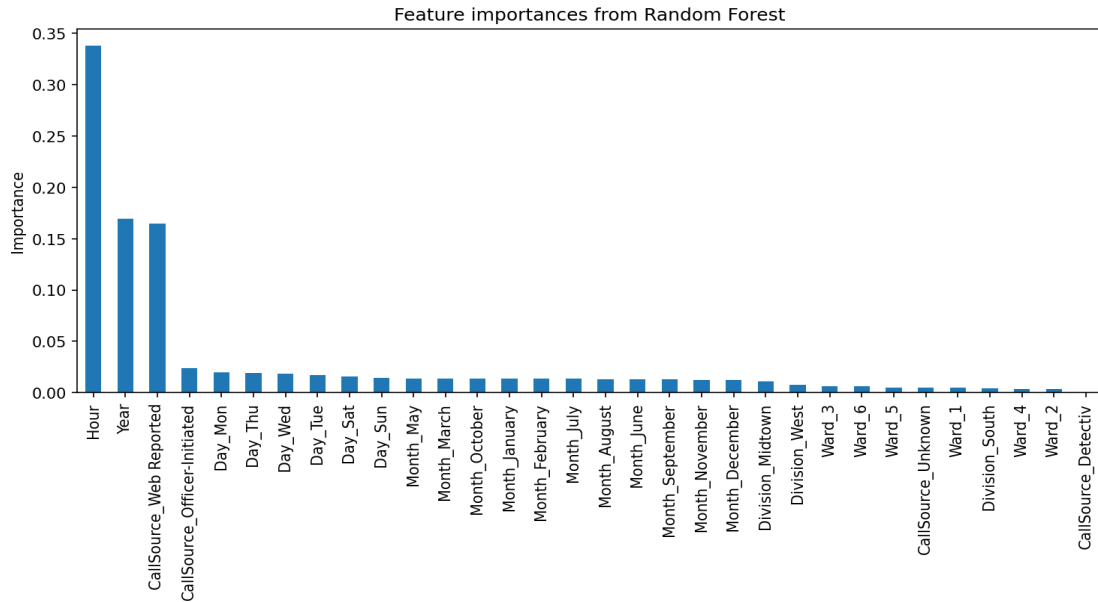


Figure 8: Feature importances from Random Forest. Hour is the strongest predictor at 0.34, followed by Year and CallSource Web Reported.

Hour is by far the most predictive feature, reflecting that different crimes tend to occur at different times of day. The high importance of CallSource Web Reported reflects that crimes reported online are overwhelmingly Larceny, giving this binary feature strong discriminative power. Year captures long-term trends including the COVID dip and post-pandemic recovery. The remaining selected features are day-of-week and month indicators, which together capture weekly and seasonal crime patterns. Notably, all Ward indicator columns had importance scores near zero, suggesting that the four broad police divisions already capture the relevant geographic signal and that finer ward-level granularity adds little predictive value.

5 Classification Models

5.1 Experimental Setup

The cleaned dataset was split into 80% training (194,501 records) and 20% test (48,626 records) using stratified sampling to preserve class proportions across both splits. All models were evaluated using macro-averaged F1 score rather than accuracy. Accuracy is misleading on imbalanced datasets: a model that predicts Larceny for every record would achieve 70% accuracy while being completely useless. Macro F1 treats all classes equally regardless of size, making it a more honest measure of performance across all 8 crime types. All models were trained with `class_weight=balanced` where available, and features were scaled using `StandardScaler` before training Logistic Regression, LDA, and SVM. Cross-validation used `StratifiedKFold` with 5 splits to ensure each fold preserves the original class proportions, which is critical given the severe class imbalance in this dataset.

5.2 Results

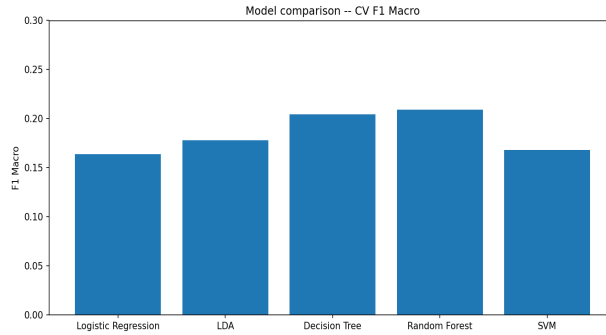


Figure 9: Cross-validated macro F1 scores for all five models. Random Forest achieves the highest score at 0.2090.

Five classifiers were trained and evaluated: Multinomial Logistic Regression, Linear Discriminant Analysis, Decision Tree, Random Forest, and Support Vector Machine with an RBF kernel. Random Forest achieved the highest 5-fold cross-validated macro F1 score (0.2090 +/- 0.0018), followed by Decision Tree (0.2045 +/- 0.0025), LDA (0.1780 +/- 0.0011), SVM (0.1683 +/- 0.0006), and Logistic Regression (0.1637 +/- 0.0011). The consistently low standard deviations confirm these results are stable across folds and not artifacts of a particular split.

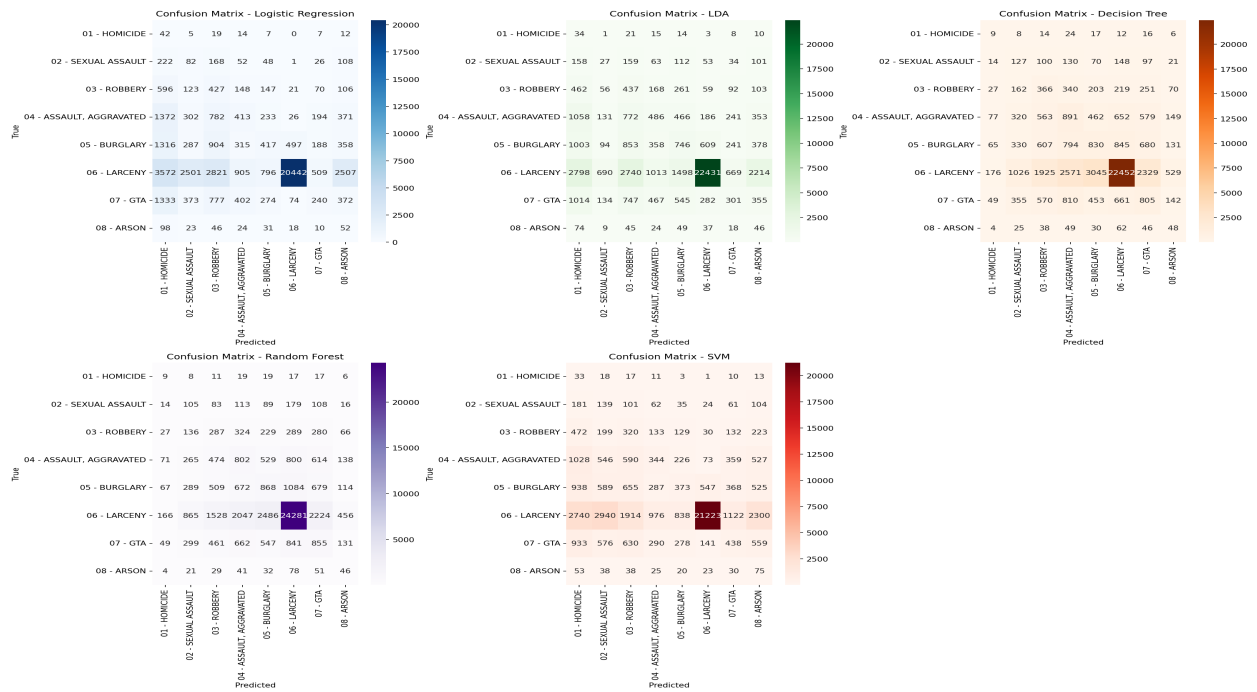


Figure 10: Confusion matrices for all five models. Larceny is predicted most accurately; Homicide, Arson, and Sexual Assault are consistently misclassified.

The confusion matrices reveal a consistent pattern across all models: Larceny is identified with reasonable accuracy while the rarest classes (Homicide, Arson, Sexual Assault) are almost always misclassified into larger categories. This persists even with balanced class weights, reflecting the fundamental difficulty of distinguishing rare crime types using only temporal and geographic features.

5.3 Individual Model Results

Logistic Regression Classification Report				
	precision	recall	f1-score	support
01 - HOMICIDE	0.0	0.4	0.01	106.0
02 - SEXUAL ASSAULT	0.02	0.12	0.04	707.0
03 - ROBBERY	0.07	0.26	0.11	1638.0
04 - ASSAULT, AGGRAVATED	0.18	0.11	0.14	3693.0
05 - BURGLARY	0.21	0.1	0.13	4282.0
06 - LARCENY	0.97	0.6	0.74	34053.0
07 - GTA	0.19	0.06	0.09	3845.0
08 - ARSON	0.01	0.17	0.02	302.0
accuracy	0.45	0.45	0.45	0.45
macro avg	0.21	0.23	0.16	48626.0
weighted avg	0.73	0.45	0.55	48626.0

Figure 11: Logistic Regression. Macro F1 = 0.1637.

LDA Classification Report				
	precision	recall	f1-score	support
01 - HOMICIDE	0.01	0.32	0.01	106.0
02 - SEXUAL ASSAULT	0.02	0.04	0.03	707.0
03 - ROBBERY	0.08	0.27	0.12	1638.0
04 - ASSAULT, AGGRAVATED	0.19	0.13	0.15	3693.0
05 - BURGLARY	0.2	0.17	0.19	4282.0
06 - LARCENY	0.95	0.66	0.78	34053.0
07 - GTA	0.19	0.08	0.11	3845.0
08 - ARSON	0.01	0.15	0.02	302.0
accuracy	0.5	0.5	0.5	0.5
macro avg	0.21	0.23	0.18	48626.0
weighted avg	0.71	0.5	0.59	48626.0

Figure 12: LDA. Macro F1 = 0.1780.

Decision Tree Classification Report				
	precision	recall	f1-score	support
01 - HOMICIDE	0.02	0.08	0.03	106.0
02 - SEXUAL ASSAULT	0.05	0.18	0.08	707.0
03 - ROBBERY	0.09	0.22	0.13	1638.0
04 - ASSAULT, AGGRAVATED	0.16	0.24	0.19	3693.0
05 - BURGLARY	0.16	0.19	0.18	4282.0
06 - LARCENY	0.9	0.66	0.76	34053.0
07 - GTA	0.17	0.21	0.19	3845.0
08 - ARSON	0.04	0.16	0.07	302.0
accuracy	0.52	0.52	0.52	0.52
macro avg	0.2	0.24	0.2	48626.0
weighted avg	0.67	0.52	0.58	48626.0

Figure 13: Decision Tree. Macro F1 = 0.2045.

Random Forest Classification Report				
	precision	recall	f1-score	support
01 - HOMICIDE	0.02	0.08	0.04	106.0
02 - SEXUAL ASSAULT	0.05	0.15	0.08	707.0
03 - ROBBERY	0.08	0.18	0.11	1638.0
04 - ASSAULT, AGGRAVATED	0.17	0.22	0.19	3693.0
05 - BURGLARY	0.18	0.2	0.19	4282.0
06 - LARCENY	0.88	0.71	0.79	34053.0
07 - GTA	0.18	0.22	0.2	3845.0
08 - ARSON	0.05	0.15	0.07	302.0
accuracy	0.56	0.56	0.56	0.56
macro avg	0.2	0.24	0.21	48626.0
weighted avg	0.66	0.56	0.6	48626.0

Figure 14: Random Forest. Macro F1 = 0.2090.

SVM Classification Report				
	precision	recall	f1-score	support
01 - HOMICIDE	0.01	0.31	0.01	106.0
02 - SEXUAL ASSAULT	0.03	0.2	0.05	707.0
03 - ROBBERY	0.08	0.2	0.11	1638.0
04 - ASSAULT, AGGRAVATED	0.16	0.09	0.12	3693.0
05 - BURGLARY	0.2	0.09	0.12	4282.0
06 - LARCENY	0.96	0.62	0.76	34053.0
07 - GTA	0.17	0.11	0.14	3845.0
08 - ARSON	0.02	0.25	0.03	302.0
accuracy	0.47	0.47	0.47	0.47
macro avg	0.2	0.23	0.17	48626.0
weighted avg	0.72	0.47	0.56	48626.0

Figure 15: SVM (RBF kernel). Macro F1 = 0.1683.

Logistic Regression achieves near-zero F1 for Homicide and Arson despite balanced weighting, reflecting the near-impossibility of predicting these classes from the available features alone. LDA improves slightly, likely benefiting from its shared covariance assumption on this dataset. Decision Tree and Random Forest show meaningful gains by capturing non-linear feature interactions; Random Forest reduces overfitting through ensembling to achieve the best overall class balance. SVM with an RBF kernel performs comparably to LDA; its non-linear boundary capability is constrained here by the weak separability of crime types in feature space. The overall performance across all models is modest, with macro F1 scores ranging from 0.16 to 0.21, reflecting a fundamental limitation: time of day, day of week, geographic division, and call source capture when and where crimes occur but not the circumstances that determine what type of crime it is.

6 Clustering and PCA

6.1 Principal Component Analysis

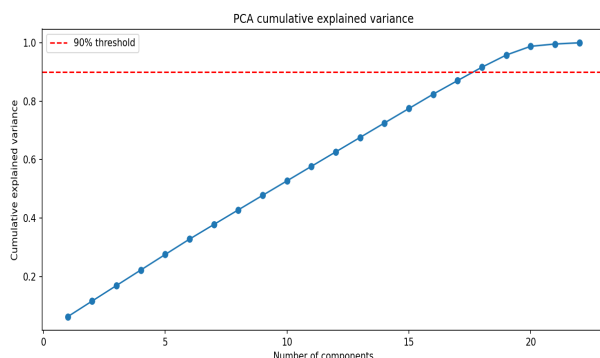


Figure 16: Cumulative explained variance. Approximately 19 components reach 90%.

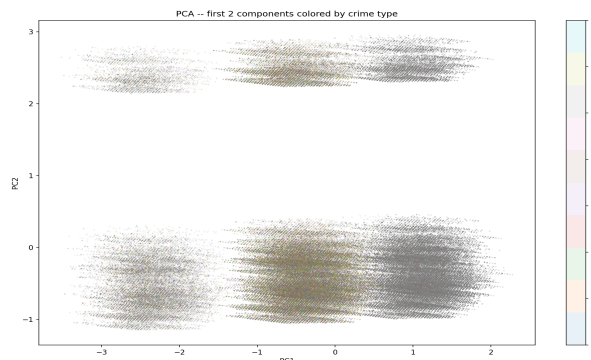


Figure 17: First two PCA components colored by crime type. Two bands visible but types are mixed within each.

Principal Component Analysis was applied to the scaled 22-feature matrix to explore the underlying structure of the data. The cumulative variance curve is nearly linear with no sharp elbow, indicating that no single feature or small group of features dominates the variance; approximately 19 components are needed to explain 90% of the total variance. The 2D PCA plot reveals two distinct horizontal bands driven by the CallSource feature: Web Reported crimes (predominantly Larceny) form one band while crimes reported through all other channels form another. Within each band, crime types are completely mixed with no visible separation, confirming that crime types do not occupy distinct regions in the feature space and is consistent with the modest classification performance observed in Section 5.

6.2 K-Means Clustering

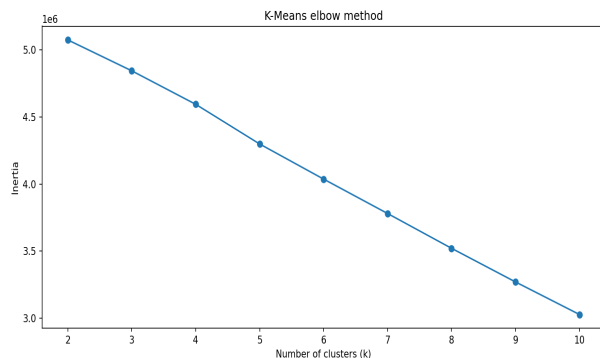


Figure 18: K-Means elbow plot. Inertia decreases nearly linearly with no clear elbow.

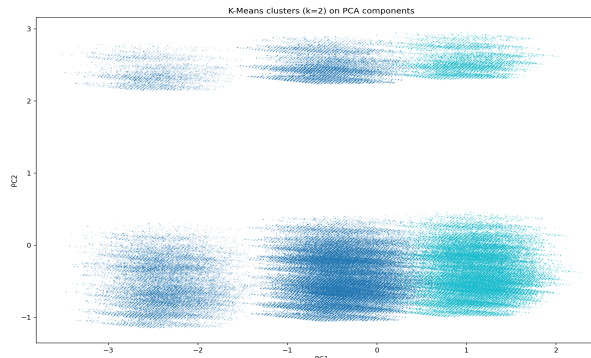


Figure 19: K-Means clusters ($k=2$) on PCA components. Split occurs along PC1.

The elbow plot shows no clear optimal k , indicating no strong natural cluster structure in the data. Despite this, $k=2$ was selected based on the two horizontal bands observed in the PCA visualization. K-Means split the data along PC1 into Cluster 0 (137,730 records) and Cluster 1 (105,397 records).

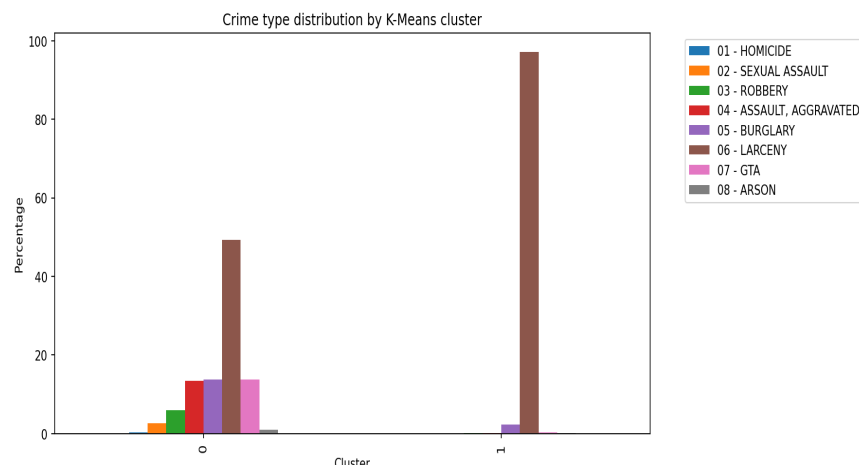


Figure 20: Crime type distribution by K-Means cluster. Cluster 1 is composed of approximately 97% Larceny records.

The most striking finding from the unsupervised analysis is that Cluster 1 is composed of approximately 97% Larceny records while Cluster 0 contains a diverse mix of all crime types. K-Means independently discovered the same dominant pattern found in supervised learning: Larceny is fundamentally distinct from all other crime types in its temporal and geographic footprint. This convergence between supervised and unsupervised results provides strong evidence that this distinction is a genuine structural feature of the data.

6.3 Hierarchical Clustering

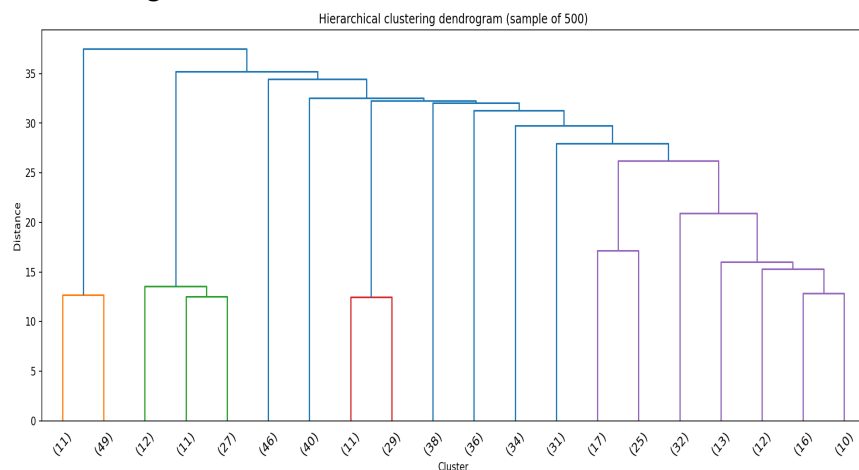


Figure 21: Hierarchical clustering dendrogram on a sample of 500 records using Ward linkage. No obvious cut point is visible.

Hierarchical clustering on a random sample of 500 records using Ward linkage produces a dendrogram with fairly uniform merge distances and no obvious cut point suggesting a natural number of clusters. This is consistent with the K-Means elbow plot and confirms that the available features do not produce strongly separated natural groupings beyond the Larceny versus non-Larceny distinction already identified.

7 Findings, Recommendations, and Conclusions

7.1 Key Findings

This project applied a full statistical machine learning pipeline to 243,127 Tucson crime records spanning 2018 through 2025. Class imbalance is the dominant challenge: Larceny accounts for approximately 70% of records while Homicide represents only 0.2%, making accurate prediction of minority crime types extremely difficult even with balanced class weights. The available features capture when and where crimes occur but not the circumstances that drive crime type, which limits classification performance. Random Forest achieved the best macro F1 (0.2090), followed by Decision Tree (0.2045), LDA (0.1780), SVM (0.1683), and Logistic Regression (0.1637).

Larceny has a fundamentally distinct pattern from all other crime types, a finding that emerged independently from both supervised and unsupervised analyses. In classification it consistently achieved the highest per-class F1; in K-Means clustering one cluster was composed of approximately 97% Larceny records; and in PCA the two horizontal bands in the 2D visualization are explained by Larceny's distinct Web Reported call source pattern during daytime hours. COVID-19 also had a measurable impact on crime volume, with counts dropping noticeably in 2020 and 2021 and Larceny showing the largest absolute decline, likely reflecting reduced retail activity during lockdowns.

7.2 Recommendations

Better features are the most impactful next step. The current set captures temporal and geographic context but not the circumstances of the crime; adding proximity to commercial areas, population density by ward, or local weather data could meaningfully improve classification accuracy. Point-level geographic coordinates would further enable spatial clustering and hotspot analysis beyond what Division and Ward allow.

A dedicated binary classifier predicting whether an incident is Larceny could achieve high accuracy and have direct operational value for resource allocation. Additionally, techniques such as SMOTE could improve recall for rare classes like Homicide and Arson beyond what balanced class weights alone can achieve.

7.3 Conclusions

This project demonstrated the application of a complete statistical machine learning pipeline from data cleaning and exploratory analysis through supervised classification and unsupervised clustering on a real-world public safety dataset. While predictive performance was modest, the analysis yielded meaningful insights about crime patterns in Tucson. The dominance of Larceny, the measurable impact of COVID-19 on crime volume, the importance of hour of day as a predictor, and the absence of strong natural cluster structure beyond the Larceny separation all contribute to a richer understanding of crime in the city. The results also highlight a fundamental lesson: model performance is constrained by the quality and relevance of the available features, and richer data would make these methods significantly more powerful.

References

- [1] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12.
- [2] Breiman, L. (2001). Random Forests. *Machine Learning*, 45.
- [3] Tucson Police Department. (2025). Reported Crimes Public Dashboard. <https://policeanalysis.tucsonaz.gov/pages/reported-crimes>
- [4] Project Repository: <https://github.com/viswakaruturi/tucson-crime-insights>
- [5] Interactive Dashboard: <https://tucson-crime-insights.streamlit.app/>